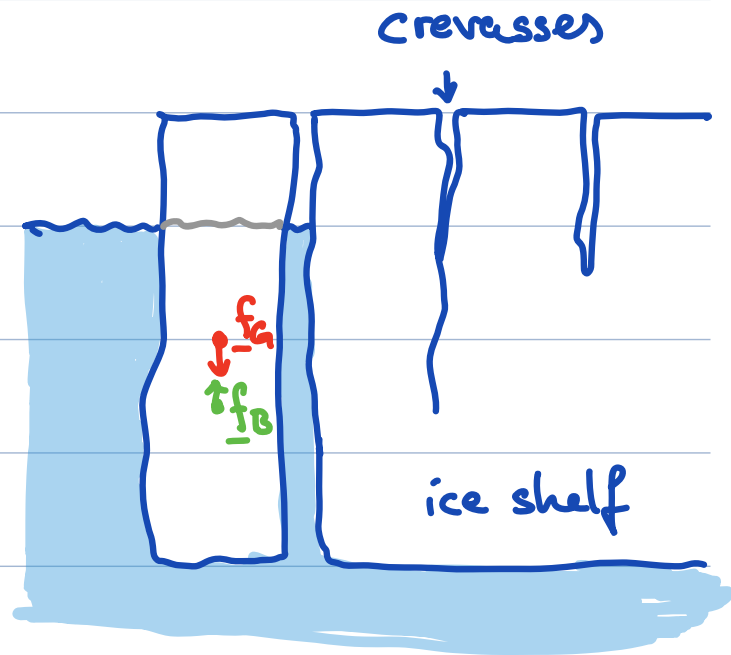


Stability calving fronts

Ice bergs formed by calving are often close to rectangular prisms and metastable.



Rectangular Prism

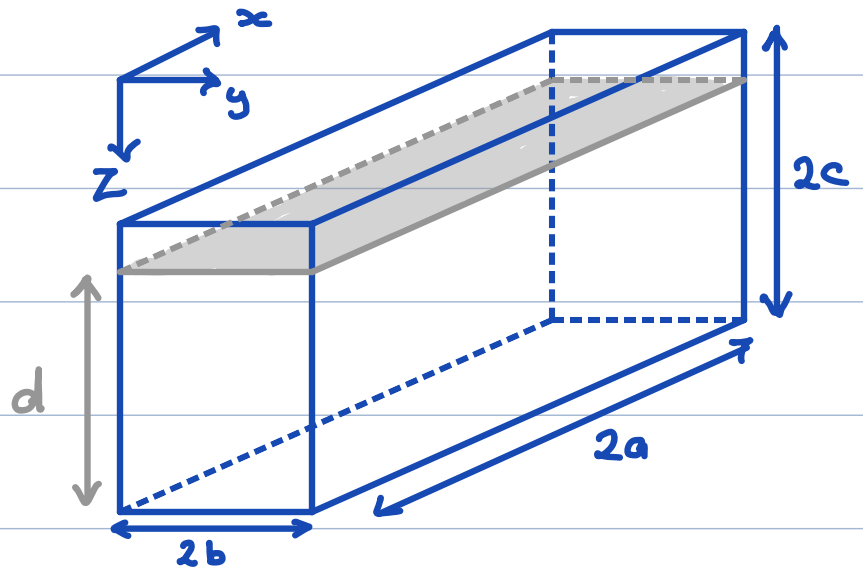
dimension: $2a, 2b, 2c$

draft: d

Total volume: $V_T = 8abc$

Displaced volume: $V_D = 4abd$

Waterline area: $A = 4ab$

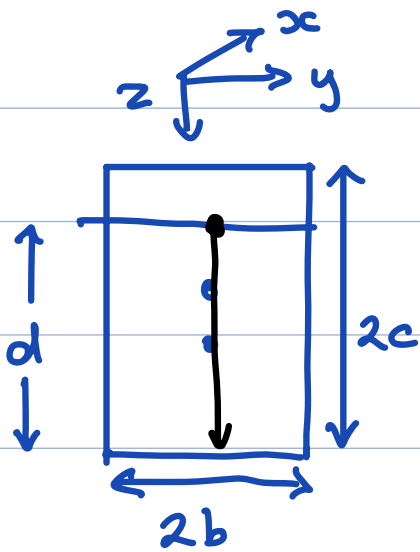


Stability to rotation around
 x -axis $\Rightarrow$ yz -cross section

$\Rightarrow$ GEO 371C Glaciology (Catania)

Iceberg Stability

Initial Geometry



2D volume / Areas:

Total: $A_T = 4bc$

Buoyancy: $A_B = 2bd$

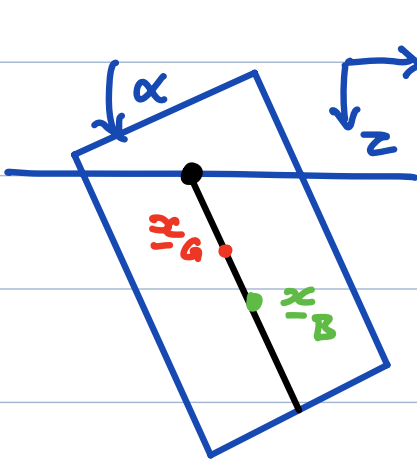
Centroids:

Gravity: $x_G = y_G = 0 \quad z_G = d - c > 0$

Buoyancy: $x_B = y_B = 0 \quad z_B = c > 0$

$\Rightarrow$ meta stable

Rotated Geometry:



counter clockwise

around x-axis: $\alpha > 0$

hydrostatic torque:

$$\underline{\tau}_H = m(\underline{x}_G - \underline{x}_B) \times \underline{g} \quad \underline{g} = -g\hat{z}$$

$$\underline{a} \times \underline{g} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ a_x & a_y & a_z \\ 0 & 0 & -g \end{vmatrix} = \hat{x}(-ga_y - 0) - \hat{y}(-ga_x - 0) + \hat{z}(0 - 0)$$

$$\underline{a} \times \underline{g} = -a_y g \hat{x} + a_x g \hat{y}$$

substitute into $\underline{\tau}_H$:

$$\tau_x = -(y_G - y_B) mg$$

rotating around $\hat{x}$

$$\tau_y = +(x_G - x_B) mg$$

$$x_G = x_B = 0$$

$$\tau_z = 0$$

$$\Rightarrow \tau_x = -(y_G - y_B) mg$$

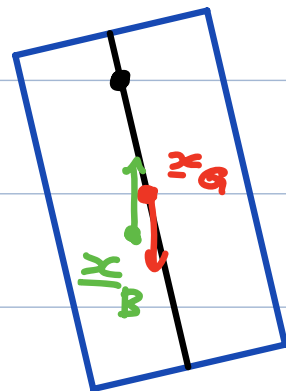
stabilizing: $\tau_x < 0$ (clockwise)

destabilizing: $\tau_x > 0$ (counterclockwise)

(assumes $\alpha > 0$ counterclockwise)

Stability criterium: ($m > 0, g > 0$)

$$\frac{\tau_x}{mg} = y_B - y_G < 0$$



horizontal displacement
of center of gravity
exceeds center of buoyancy

Horizontal displacements

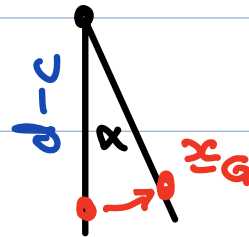
assume $\alpha \ll 1 \Rightarrow \tan \alpha \approx \sin \alpha \approx \alpha$

1) Center of gravity

shape of ice berg remains same

$\Rightarrow$ simple rotation

$$y_G \approx \alpha(d-c) > 0$$



2) Center of buoyancy

shape of submerged body changes

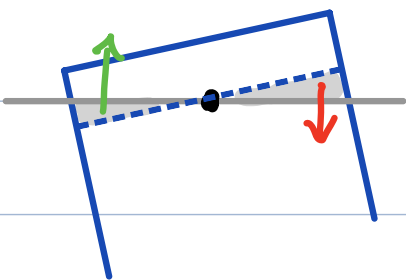
y_B changes due to

1) rotation Δy_B^R

2) change in shape Δy_B^S

$$\Rightarrow y_B = \Delta y_B^R + \Delta y_B^S$$

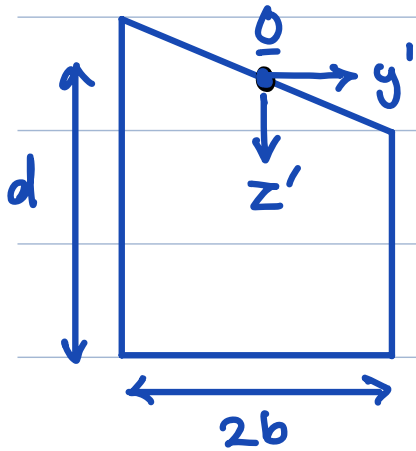
as before: $\Delta y_B^R \approx \alpha \frac{d}{2}$ destabilizing ($z_B > z_G$)



change in shape is stabilizing!

Lateral change due to change in shape

new coordinate system y' & z'
aligned with trapezoid



$$A_B = 2bd$$

for $\alpha \ll 1$: $y' \approx y$

$$z'_{\text{Top}} = \alpha y'$$

$$z'_{\text{Bot}} = d$$

y-centroid:
$$\Delta y_B^S = \frac{1}{A_B} \int_{-b}^b \int_{z'_{\text{Top}}}^{z'_{\text{Bot}}} y' dz' dy'$$

$$\int_{z'_{\text{Top}}}^{z'_{\text{Bot}}} dz' = z'_{\text{Bot}} - z'_{\text{Top}} = d - \alpha y'$$

substitute:
$$\Delta y_B^S = \frac{1}{A_B} \int_{-b}^b y' (d - \alpha y') dy'$$

$$= \frac{1}{A_B} \left(\frac{dy'^2}{2} - \frac{\alpha y'^3}{3} \right) \Big|_{-b}^b = -\frac{1}{A_B} \frac{2\alpha b^3}{3}$$

$$A_B = 2bd \quad \Delta y_B^S = -\frac{\alpha b^2}{3d} \Rightarrow \text{stabilizing}$$

Vertical centroid:
$$\frac{d}{2} - \frac{\alpha^2 b^2}{6d} \approx \frac{d}{2} \Rightarrow \text{no change!}$$

Total horizontal displacement:

$$y_B = \Delta y_B^R + \Delta y_B^S \\ = \frac{\alpha d}{2} - \frac{\alpha b^2}{3d}$$

Substitute into stability criterium:

$$\frac{T_x}{mg} = y_B - y_G < 0 \quad y_G = \alpha(d-c)$$

$$\cancel{\frac{\alpha d}{2}} - \cancel{\frac{\alpha b^2}{3d}} - \cancel{\alpha}(d-c) < 0 \\ d^2 - 2cd > -\frac{2}{3}b^2$$

aspect ratios of submerged body: $\frac{b}{c}$ & $\frac{d}{c}$

$$\left(\frac{d}{c}\right)^2 - 2\frac{d}{c} > -\frac{2}{3}\left(\frac{b}{c}\right)^2$$

complete square: $\left(\frac{d}{c}\right)^2 - 2\frac{d}{c} + 1 = \left(\frac{d}{c} - 1\right)^2$

$$\boxed{\left(\frac{d}{c} - 1\right)^2 > 1 - \frac{2}{3}\left(\frac{b}{c}\right)^2}$$

$$\text{lhs: } \left[\left(\frac{d}{c}\right)^2 - 1\right]^2 \geq 0$$

rhs < 0: always stable

$$\Rightarrow 1 - \frac{2}{3} \left(\frac{b}{c} \right)^2 < 0$$

$$\frac{b}{c} > \sqrt{\frac{3}{2}}$$

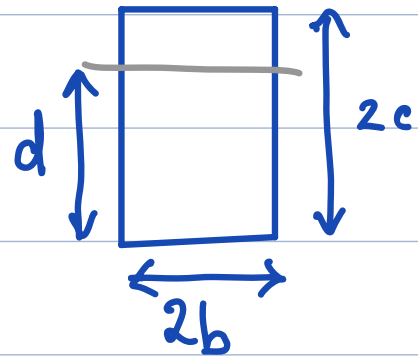
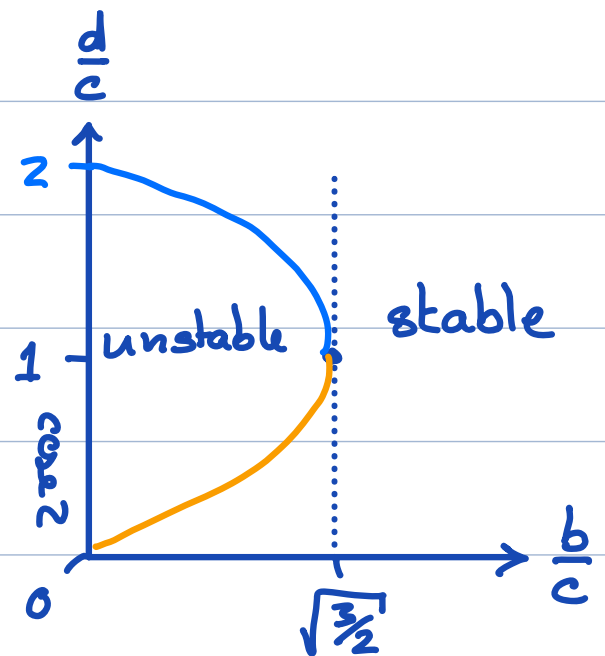
rhs > 0: cond. stable

case 1: $d/c > 1$

$$d/c > 1 + \sqrt{1 - \frac{2}{3} \left(\frac{b}{c} \right)^2}$$

case 2: $d/c < 1$

$$d/c < 1 - \sqrt{1 - \frac{2}{3} \left(\frac{b}{c} \right)^2}$$



Relate $\frac{d}{c}$ to density contrast using isostasy

$$\rho_i = 917 \frac{\text{kg}}{\text{m}^3} \quad \rho_w = 1025 \frac{\text{kg}}{\text{m}^3}$$

$$m_i = m_w$$

$$\rho_i \cancel{8a} b c = \rho_w \cancel{4a} b d \quad 2\rho_i c = \rho_w d$$

$$\Rightarrow 2\frac{d}{c} = \frac{\rho_i}{\rho_w} = 0.89 \Rightarrow 90\% \text{ under water}$$

Solve for stable aspect ratio $\frac{b}{c}$

$$\left(\frac{d}{c} - 1\right)^2 = 1 - \frac{2}{3}\left(\frac{b}{c}\right)^2$$

$$\frac{2}{3}\left(\frac{b}{c}\right)^2 = 1 - \left(\frac{d}{c} - 1\right)^2$$

$$\begin{aligned}\left(\frac{b}{c}\right)^2 &= \frac{3}{2} \left[1 - \left(\frac{d}{c} - 1\right)^2 \right] = \\ &= \frac{3}{2} \left[1 - \left(\left(\frac{d}{c}\right)^2 - 2\frac{d}{c} + 1\right) \right] \\ &= \frac{3}{2} \left[\left(\frac{d}{c}\right)^2 + 2\frac{d}{c} \right]\end{aligned}$$

$$\frac{b}{c} = \sqrt{\frac{3}{2} \left[\left(\frac{d}{c}\right)^2 + 2\frac{d}{c} \right]}$$

substituting values of ice & water $\frac{d}{c} = 1.8$

$$\frac{b}{c} \approx 0.75 = \frac{3}{4}$$

$$\frac{b}{c} \approx \frac{3}{4}$$